

Introduction to Analysis Homework 8

April, 24, 2025

1. (12 pt) Given a convergent series $\sum_{n=1}^{\infty} a_n$, where each $a_n \geq 0$. Prove that $\sum_{n=1}^{\infty} \sqrt{a_n} n^{-p}$ converges if $p > \frac{1}{2}$. Given a counterexample for $p = \frac{1}{2}$.
2. (a) (7 pt) Given that $\sum_{n=1}^{\infty} a_n$ converges, where each $a_n > 0$. Prove that $\sum_{n=1}^{\infty} (a_n a_{n+1})^{\frac{1}{2}}$ also converges.
(b) (7 pt) Given a monotonic, positive sequence $\{a_n\}$. If $\sum_{n=1}^{\infty} (a_n a_{n+1})^{\frac{1}{2}}$ converges, show that $\sum_{n=1}^{\infty} a_n$ converges.
3. Given a positive sequence $\{x_k\}$. Prove or disprove the following statements.
 - (a) (6 pt) If $\lim_{k \rightarrow \infty} x_k^{\frac{1}{k}} = l$, then $\lim_{k \rightarrow \infty} \frac{x_{k+1}}{x_k} = l$.
 - (b) (6 pt) If $\limsup_{k \rightarrow \infty} \frac{x_{k+1}}{x_k} > 1$, then $\sum_{k=1}^{\infty} x_k$ diverges.
4. (12 pt) Let $\{a_n\}$ be a sequence of positive real numbers. Show that

$$\liminf_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} \leq \liminf_{n \rightarrow \infty} a_n^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} a_n^{\frac{1}{n}} \leq \limsup_{n \rightarrow \infty} \frac{a_{n+1}}{a_n}.$$